Polynomial Filter Diagonalization and Quantum Many-Body Scars in the PXP Model

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There will be some abstract here...

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I. INTRODUCTION

Thermalization process of quantum states is a topic of central attention in quantum many body physics. One particular instance of work on this topic is the *Eigenstate Thermalization Hypothesis* (ETH), which says that although we have unitary time evolution in a closed quantum system, local observables may approach stationary values that can be described well by statistical mechanics. Therefore, a generic (non-integrable) quantum system energy eigenstate may already give us thermal expectation values for local observables [1, 2].

However, there are quantum systems that have ETH violating states. The particular example we study in this case is the *PXP Hamiltonian*, which is used to effectively describe a one dimensional *Rydberg atom chain*. Atoms in the Rydberg atom chain have two states, with $|0\rangle$ being their electronic ground state, and $|1\rangle$ being their highly excited Rydberg state. In the Rydberg state, the dipole moment of the corresponding atom is so large that it suppresses the nearby atoms from being simultaneously excited. This constraint is called the *Rydberg blockade*. This constraint is modeled by the PXP Hamiltonian through projected spin flip dynamics.

The ETH violating states of the PXP Hamiltonian are called the *scar states*. Scar states are characterized by their unusually low entanglement compared to other eigenstates at similar energies. They also have large overlap with some initial states, called the *Néel type* states, which gives us persistent many body revivals in the time evolution of the system [2]. They only form a small part of the spectrum, and they are evenly spaced within the whole spectrum, which makes pinpointing the scar states a numerically interesting problem.

Like with any state finding problem, the problem gets exponentially difficult with increasing system size L, this case as 2^L. In this work, our first step was to restrict the system to the blockade constrained Hilbert space by analyzing the adjacency graph of the Hamiltonian, and decomposing it to its momentum sectors using the translation invariance of the system, while using the exact diagonalization of $L = 16$ case as benchmark. The entanglement entropy can be then calculated using simple reshaping of the eigenstate and applying a singular value decomposition (SVD). Lastly, Chebyshev polynomial filtering was used to determine the scar states by targeting specific energies without diagonalizing the whole Hamiltonian.

II. THE PXP HAMILTONIAN

A. The Action of the PXP Hamiltonian

To study the Rydberg blockade mechanism, we introduce the PXP Hamiltonian

$$H = \sum_{i=0}^{L-1} P_i X_{(i+1)\%L} P_{(i+2)\%L}, \tag{1}$$

where $X_i = \mathbb{1} \otimes \sigma_x \otimes \mathbb{1}$ is the Pauli matrix $X = \sigma_x$, acting on site i, and the projector $P_i = \frac{1}{2}(\mathbb{1} - Z_i)$ is described in terms of the identity $\mathbb{1}$ and the Pauli matrix $Z = \sigma_z$, which acts on site i. The Hilbert space $\mathcal{H}$ is defined such that $Z|0\rangle = -|0\rangle$ and $Z|1\rangle = |1\rangle$, such that $P|1\rangle = 0$ and $P|0\rangle = |0\rangle$. For a computational basis state

$$|n_0 n_1 \cdots n_{L-1}\rangle, \qquad n_i \in \{0, 1\},$$

the Hamiltonian acts by checking whether the two neighboring sites i and $i+2$ are both in the state $|0\rangle$. If they are, the operator flips the middle spin at site $i+1$, if they are not, the term gives zero. A term in the Hamiltonian can be seen explicitly as,

$$P_i X_{(i+1)\%L} P_{(i+2)\%L} |\cdots n_i n_{i+1} n_{i+2} \cdots\rangle = \begin{cases} |\cdots 0(n_{i+1} \oplus 1)0 \cdots\rangle, & n_i = 0,\ n_{i+2} = 0, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Here, the $\oplus$ in Equation (2) is used to emphasize that the summation is in modulo 2.

To see the action explicitly, let us consider the local

three-site basis

$$|n_0 n_1 n_2\rangle\,, \qquad n_i \in \{0,1\}.$$

If we take one of the local Hamiltonian terms $h_0 = P_0 X_1 P_2$, we can see that h_0 acts only when both neighboring sites of n_1 are in the state $|0\rangle$. Therefore, it maps two of the basis states as

$$h_0\,|000\rangle = |010\rangle\,,$$
$$h_0\,|010\rangle = |000\rangle\,,$$

while the rest of the $h_0\,|n_0 n_1 n_2\rangle = 0$. Thus, a spin can only flip if both of its neighboring sites are in the state $|0\rangle$. This effectively models the blockade constraint.

B. Blockade Constrained Hilbert Space

The Rydberg blockade constraint forbids both neighboring atoms to be in the excited state at the same time. This means the states that contain adjacent $|\cdots 11\cdots\rangle$ configurations are not physically relevant. If we look at the adjacency graph of the Hamiltonian, which is constructed by connecting two basis states whenever the corresponding Hamiltonian matrix element $\langle n_0 n_1 \cdots|\, H\, |n_0' n_1' \cdots\rangle$ is nonzero, we see that the largest connected component is formed by the basis states that satisfy the Rydberg blockade constraint. Therefore, we restrict ourselves to the biggest block of the Hamiltonian for our calculations.

Another curious property of the biggest block is its dimension. Assume we have a chain of atoms of size L, and we have open boundary conditions, then we have the Hilbert space of the biggest block with dimension

$$\dim \mathcal{H}_{\text{biggest}}(L) = \mathcal{D}(L), \tag{3}$$

and we add one more atom to our basis state. An atom has the states $|0\rangle$ and $|1\rangle$. The state $|0\rangle$ of the new atom can be freely attached to any biggest block basis state in the L-atom chain, giving us $\mathcal{D}(L)$ number of valid configurations. When adding the state $|1\rangle$ of the new atom to the L-atom chain, the Lth atom must be in the state $|0\rangle$, and we can again attach this pair $|0\rangle_L\,|1\rangle_{L+1}$ freely to any biggest block basis state in the remaining $(L-1)$-atom chain, giving us $\mathcal{D}(L-1)$ number of valid configurations. Therefore, the new $(L+1)$-atom chain has the Hilbert space dimension

$$\mathcal{D}(L+1) = \mathcal{D}(L) + \mathcal{D}(L-1), \tag{4}$$

which implies a Fibonacci sequence, with the initial conditions $\mathcal{D}(1) = 2$ and $\mathcal{D}(2) = 3$. This also implies that the dimension of the biggest block Hilbert space $\mathcal{D}(L)$ of the PXP Hamiltonian grows with the golden ratio

$$\phi^L = \left(\frac{1+\sqrt{5}}{2}\right)^L \approx (1.618)^L,$$

with growing system sizes, rather than 2^L.

For periodic boundary conditions, the blockade constraint also acts between the first and last atoms, which changes the counting from the open-chain case. This time, we can split the allowed configurations according to the value of the first site. If $|n_0\rangle = |0\rangle$, the remaining $L-1$ sites form an open chain with the blockade constraint, giving $\mathcal{D}_{\text{open}}(L-1)$ configurations. If $|n_0\rangle = |1\rangle$, then periodicity forces both neighboring sites to be zero, $|n_1\rangle = |n_{L-1}\rangle = |0\rangle$. The remaining free part can then be thought of as an $(L-3)$-atom chain with open boundary conditions, which has $\mathcal{D}_{\text{open}}(L-3)$ configurations. Thus we have

$$\mathcal{D}_{\text{periodic}}(L) = \mathcal{D}_{\text{open}}(L-1) + \mathcal{D}_{\text{open}}(L-3). \tag{5}$$

Using $\mathcal{D}_{\text{open}}(L) = F_{L+2}$, Equation (5) becomes

$$\mathcal{D}_{\text{periodic}}(L) = F_{L+1} + F_{L-1}, \tag{6}$$

which is the L-th Lucas number [3]. The constrained Hilbert space still grows asymptotically as ϕ^L, but with the periodic boundary condition included exactly.

C. Translation Invariance and Momentum Sectors

The PXP Hamiltonian has translation symmetry, since every site sees the same rule and the chain is periodic. We can use this to create a basis of symmetrized momentum states and block diagonalize the biggest block in momentum sectors with the moment k. The construction of the momentum basis in this section follows the lecture notes [4].

To use the translation invariance in our favor, we define the translation operator T as the operator which shifts every spin by one site,

$$T\,|n_0 n_1 \cdots n_{L-1}\rangle = |n_{L-1} n_0 \cdots n_{L-2}\rangle\,. \tag{7}$$

Since the Hamiltonian is a sum over identical local terms on every site, translation does not change the Hamiltonian. Therefore,

$$[H, T] = 0.$$

This means we can diagonalize H and T simultaneously, and the Hamiltonian can be decomposed into momentum sectors with the basis of translation invariant momentum states

$$|\psi_{k,\vec{n}}\rangle = \frac{1}{\mathcal{N}_{k,\vec{n}}} \sum_{r=1}^{L} e^{ikr} T^r\,|\text{rep}(\vec{n})\rangle\,. \tag{8}$$

Here $k = 2\pi n_k/L$ are the momentum sectors, $n_k \in \mathbb{Z}_L$, and $|\text{rep}(\vec{n})\rangle$ are representatives of different families (orbits) $\mathcal{F}(\vec{n})$, created by the translation operator T, defined as

$$|\text{rep}(\vec{n})\rangle = T^{x_{\vec{n}}}\,|\vec{n}\rangle\,. \tag{9}$$

Here, we use $|\vec{n}\rangle = |n_0 n_1 \cdots n_{L-1}\rangle$ as a shorthand notation. The representative $|\text{rep}(\vec{n})\rangle$ of a family is usually chosen as the index that has the lowest binary representation.

When we look at the normalization of Equation (8) for $f_{\vec{n}} = \#\mathcal{F}(\vec{n}) < L$, the normalization condition gives us a selection rule

$$f_{\vec{n}} n_k \pmod L = 0, \tag{10}$$

because the momentum projection is not always normalizable. If this rule is not satisfied, then the projected momentum state must be omitted from the basis. For normalizable states, we can write

$$|\psi_{k,\vec{n}}\rangle = \frac{1}{\sqrt{f_{\vec{n}}}} \sum_{r=1}^{f_{\vec{n}}} e^{ikr} T^r \, |\text{rep}(\vec{n})\rangle \, . \tag{11}$$

III. NUMERICAL METHODS and RESULTS

A. Sparse Hamiltonian and Adjacency Graph

A sparse matrix is a matrix where there are much more zero elements than nonzero elements. We can conveniently store the information of such matrices by only considering their nonzero elements. The matrix form of the PXP Hamiltonian is also sparse (refer to Equation (A1) for an example for $L = 4$), and we can find its elements by the algorithm,

1. Generate the computational basis states $\{|i\rangle\}$.
2. Initialize empty lists for the row indices, column indices, and matrix entries.
3. For each basis state $|i\rangle$ and each lattice site n, apply the local operator $P_{n-1} X_n P_{n+1}$.
4. If the action is allowed, identify the resulting state $|j\rangle$. We now have $\langle j| H |i\rangle = H_{ji}$.
5. Append j to the row list, i to the column list and a 1 to the data list.
6. Construct the sparse Hamiltonian by feeding the lists to `scipy.sparse.csr_matrix`.

After this, the adjacency graph can be conveniently drawn using `networkx.from_scipy_sparse_array`. The examples for full matrix and its adjacency graph can be seen in Equation (A1) and Figure 7 respectively.

B. Exact Diagonalization Benchmark

To be used as benchmark, we fully diagonalize the biggest block of the PXP Hamiltonian by using the dense matrix version of the Hamiltonian. Different blocks of the Hamiltonian can be accessed by using `networkx.connected_components` after forming the adjacency matrix. The biggest block of the Hamiltonian is then accessed by taking the list which has the longest length in a list of lists we get from `networkx.connected_components`. The exact diagonalization is then carried out by `scipy.linalg.eigh` after converting the sparse matrix to array. The result is then stored to be used as benchmark. The full dense spectrum for $L = 16$ case can be seen in Figure 1.

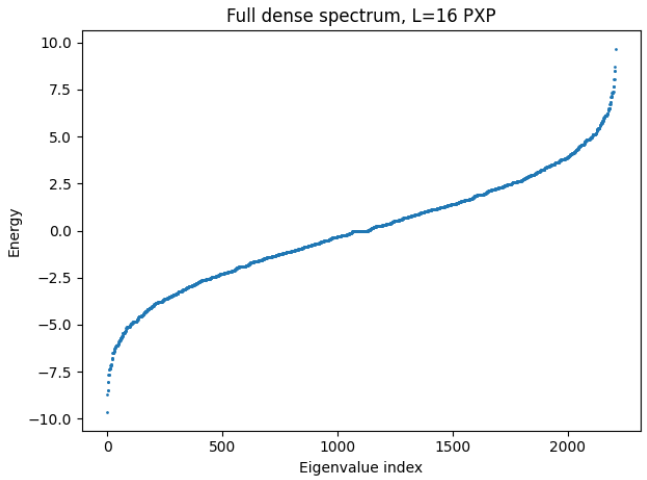


FIG. 1. Full dense spectrum for the $L = 16$ PXP Hamiltonian.

C. Momentum Sector Basis Construction

Momentum sector basis construction can be straightforwardly done using Equation (11). First, the sparse Hamiltonian is constructed with the related algorithm in Section III A. Then, the biggest block is accessed by using `networkx.connected_components` and looking at the block with the maximum length. After this, the basis states that belogn to H_{biggest} are grouped into translation orbits (families), and the smallest integer index is chosen as the representative of their corresponding orbits. Not every orbit can contribute to every momentum sector, so we need to do a selection by using Equation (10) for every k. We also need to be careful using only the basis states inside H_{biggest} when we want to only work with the basis of H_{biggest} rather than H.

The resulting momentum states are then stored as the columns of a transformation matrix V_k. The Hamiltonian restricted to the momentum sector k is then obtained by projection,

$$H_k = V_k^\dagger H_{\text{biggest}} V_k. \tag{12}$$

This way, we can diagonalize each H_k independently, which considerably reduces the dimensions of the matrices involved in the process. However, this straightforward way of finding the momentum sector basis becomes

the bottleneck when we are trying to scale up to larger system sizes L.

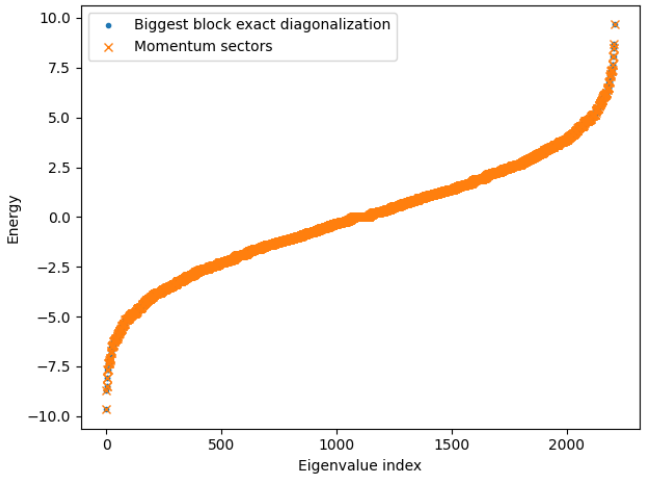


FIG. 2. Comparison of the biggest block exact diagonalization and the momentum sectors.

D. Entanglement Entropy via Reshaping

The scar states are identified by their unusually low entropy compared to usual thermal eigenstates at similar energies. Thus, comparing the entanglement entropy against the energy is a convenient way to identify the scar states inside the spectrum. The computation of the entanglement entropy follows the script by Ulrich Schollwöck [5].

To compute the entanglement entropy, the system is thought of as a bipartite system. For a chain of length L, we divide the system into two subsystems A and B, with lengths L_A and L_B, such that $L_A + L_B = L$. A general eigenstate can be written as

$$|\psi\rangle = \sum_{i_A, i_B} \psi_{i_A i_B} |i_A\rangle \otimes |i_B\rangle , \tag{13}$$

where $|i_A\rangle$ and $|i_B\rangle$ are computational basis states of the two subsystems. The coefficients of the state are then reshaped into a matrix

$$M_{i_A i_B} = \psi_{i_A i_B}. \tag{14}$$

The singular values of this matrix are the Schmidt coefficients of the bipartition. Equivalently, if

$$M = USV^\dagger, \tag{15}$$

then the diagonal entries s_α of S give the Schmidt coefficients.

The reduced density matrix of subsystem A, defined by $\rho_A = \mathrm{Tr}_B(\rho)$, has eigenvalues

$$\lambda_\alpha = s_\alpha^2, \tag{16}$$

and the von Neumann entanglement entropy is

$$S_A = -\sum_\alpha \lambda_\alpha \log \lambda_\alpha = -\sum_\alpha s_\alpha^2 \log s_\alpha^2. \tag{17}$$

In the numerical implementation, this can be directly computed by reshaping the eigenvector into a $2^{L/2} \times 2^{L/2}$ matrix, performing a singular value decomposition, and summing over the squared singular values. However, because of the floating point arithmetic, Schmidt values can get absurdly low, causing `numpy` to interpret them as zero inside the logarithm. Therefore a tolerance of $s_\alpha > 10^{-15}$ should be taken.

Since the Hamiltonian is diagonalized in the blockade constrained Hilbert space, the eigenvectors first have to be embedded back into the full computational basis before reshaping to compute their entropies correctly.

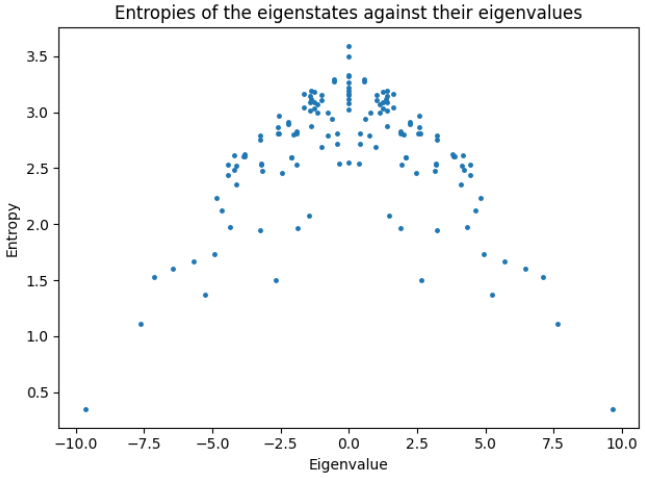


FIG. 3. Entanglement entropies of the eigenstates against their eigenvalues.

E. Chebyshev Polynomial Filtering

The scar states live on a broad spectrum, thus if we want to find them we need to be specifically targeting them by using a filtering method that amplifies the targeted energy eigenstate's eigenvalue and suppresses the rest. In theory, this can be done by using the Dirac delta function

$$\delta(E - H), \tag{18}$$

which would project onto eigenstates with energy E, so we would be able to target specific energy eigenstates directly. However, numerically this is not directly usable because the peak of Dirac delta function is infinitely sharp. Because of this, we replace the Dirac delta function with some approximation of it by expanding the delta function at the target value with some polynomial series, this time the Chebyshev polynomials.

Chebyshev polynomials are defined by

$$T_n(x) = \cos(n \arccos x), \tag{19}$$

with the inner product

$$\int_{-1}^{1} dx\, T_m(x) T_n(x) / \sqrt{1-x^2} = \delta_{nm}, \tag{20}$$

and satisfy the recursion relation

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x). \tag{21}$$

Chebyshev polynomials are used because they are the optimal way to do polynomial filtering for Hermitian matrices, and their recurrence relation gives us a convenient way to calculate the current vector product with using the old vector products [4].

After the recursive function for finding the $p(A)\vec{x}$ is defined, where A is a matrix, $p(A)$ is the polynomial filter, and $\vec{x}$ is a vector, the corresponding (Matrix $\times$ Vector) method is turned into a linear operator using `scipy.sparse.linalg.aslinearoperator`. Since the scar states of the system is not known beforehand, the system is sweeped across the eigenspectrum with target values $E_{\text{target}} = -L-1, -L, \ldots, L, L+1$ to boost the corresponding eigenvalues and suppress the rest, and for every target 20 eigenpairs were calculated using the method. There is freedom in choosing the target eigenvalue interval and the number of eigenpairs, and the chosen interval may or may not be optimal depending on system size. With this method, rather than looking at the whole eigenspectrum, which can have thousands of eigenpairs, we just look at 20 eigenpairs close to the current target energy eigenvalue.

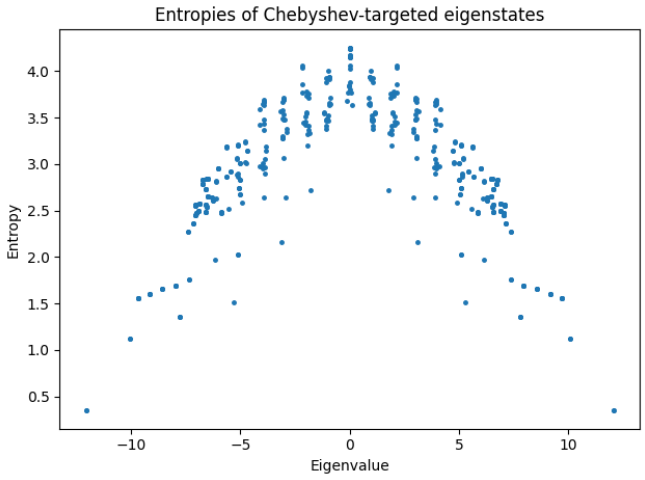


FIG. 4. Entropies of the Chebyshev-targeted eigenstates.

Since this method amplifies target energies as

$$E_{\text{target}} \longrightarrow p(E_{\text{target}}), \tag{22}$$

we need to find the actual energies once more for the entropy plot. Since the Chebyshev eigenvectors are still the same as the actual eigenvectors, i.e. we have

$$\begin{aligned} H\,|v\rangle &= E\,|v\rangle\,, \\ p(H)\,|v\rangle &= p(E)\,|v\rangle\,, \end{aligned} \tag{23}$$

we can still use these eigenvectors to find their corresponding eigenvalues by

$$\langle v|\,H\,|v\rangle = E. \tag{24}$$

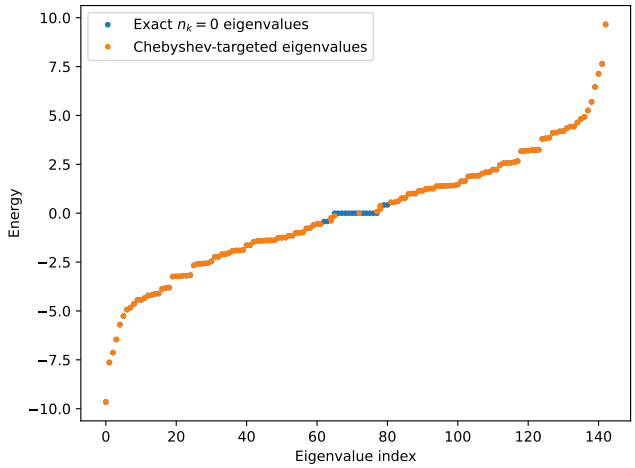


FIG. 5. Comparison of exact $n_k = 0$ eigenvalues and Chebyshev-targeted eigenvalues.

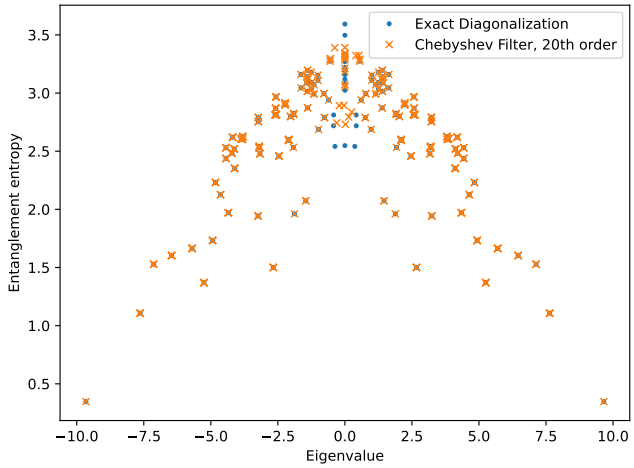


FIG. 6. Comparison of the exact diagonalization entropies and the Chebyshev filter entropies.

Appendix A: Dense Hamiltonian for $L = 4$

For illustration, the full Hamiltonian for $L = 4$ in the computational basis can be seen in Equation (A1).

$$H = \begin{pmatrix}
0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix} \tag{A1}$$

Appendix B: Adjacency Graph for $L = 4$

For illustration, the adjacency graph of thee Hamiltonian for $L = 4$ in the computational basis can be seen in Figure 7.

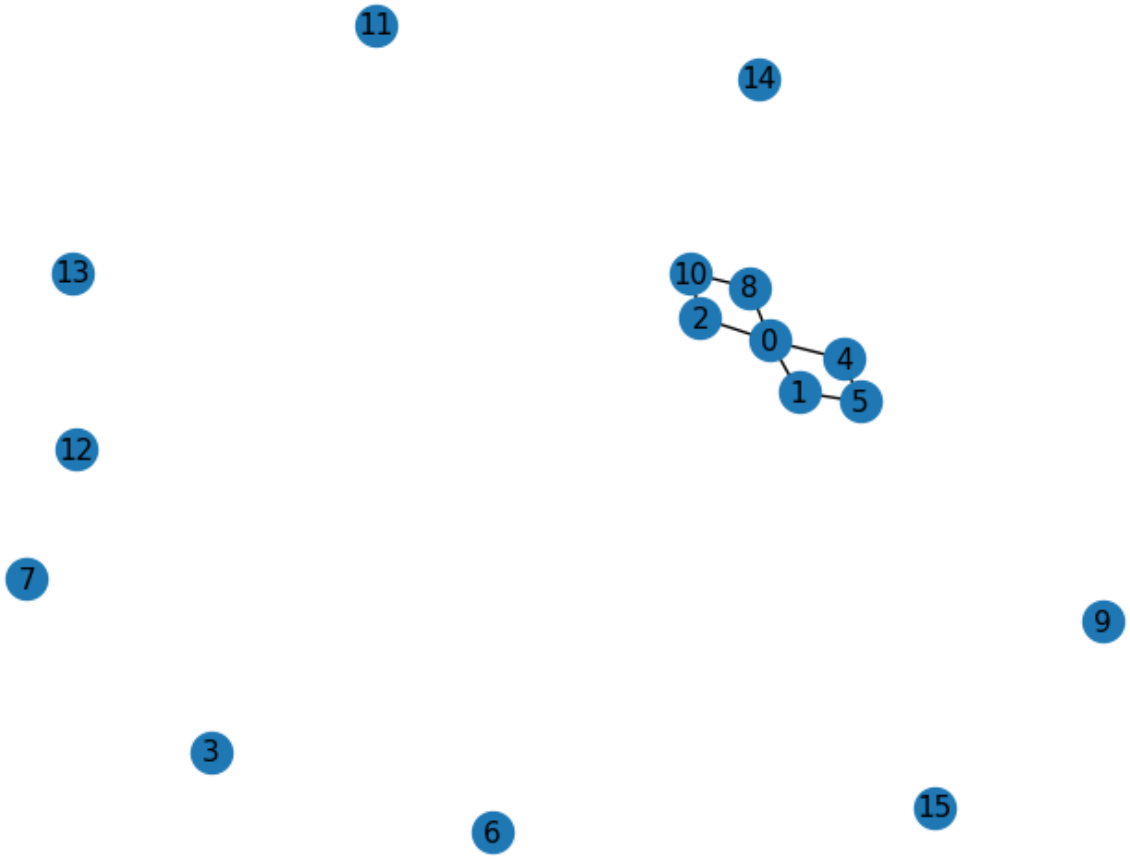


FIG. 7. The adjacency graph of thee Hamiltonian for $L = 4$

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